

HUGOPHARM TECHNOLOGIES PVT LTD	QUOTATION UNIFLUID 30-FLUID BED PROCESSOR	QUOTATION REF NUMBER HUGO/2105/2025-1 JOB NO. : - REV : 00
	CLIENT: ARHANTH BIO	SHEET NO 1 OF 11

Client: ARHANTH BIO

Contact Person: Mr. Chirag



For further assistance please contact:

Sales Office:

Hiren Panchal

HUGOPHARM TECHNOLOGIES PVT LTD

8, Jogani Industrial Estate,
541, Senapati Bapat Marg,
Dadar (W), Mumbai: 400 028.

Tel: +91-22-24226882 / Fax: +91-22-24222815 / Email: sales@sbpanchal.com

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1. UNIFLUID 30: Fluid Bed Dryer (Bulk density: 0.32gm/cc)

Sn	Item	Description
1.	<u>General Description</u>	Design and Execution process are done in with compliance to cGMP. They include zoning of processing sub assemblies from Mechanical/Electrical/Pneumatic parts. Easy, quick and comprehensive dismantling of key process parts for cleaning and /or validation. Detailed and accurate recording of operations. Integrated Compact Unit. Stainless Steel exterior. AISI 316 quality contact parts.
2.	<u>General Construction</u>	a. All the material contact parts shall be made from S.S. 316 quality stainless steel material. The bottom chamber shall be made of Stainless steel AISI 304 which connects the side of the chamber to air inlet duct. It shall be connected with drain for easy cleaning. b. Product Container shall be lifted and sealed against expansion chamber with help pneumatic cylinders. c. Expansion chamber shall be cylindrical in shape and shall be supported on side support column. It shall have sight glass and light fitting arrangement. d. Side Supports made from S.S. 304. Strategically mounted observation windows for viewing granulation process. Light fixing arrangement shall be made for fixing illumination lamp.
3.	<u>Product Container Top Spray</u>	a. Product container shall be made of Stainless Steel AISI 316, for spray granulation applications. It shall be mounted on S.S. 304 Trolley duly mounted on S.S. Bracket and twin type PU Wheels. b. It shall have an oblong shape observation window duly fitted with toughened observation glass. c. Product Container shall have product temperature sensing port and sampling port for extracting small quantities of coated substrate during operation. d. The quick remove type distribution plate & bottom mesh shall be made from S.S. 316. e. The machine shall be supplied with 2Nos of silicon molded mesh of 24x110# (Dutch Weave)

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4.	<u>Retarding Chamber cum Filter Chamber</u>	a. The retarding chamber cum filter chamber shall have provision for insertion of top spray nozzles. Three number of ports shall be provided for top spray nozzle. Blind flanges shall be provided for covering of the nozzle ports if only drying mode is desired. b. A light glass shall be fitted in the retarding chamber for illuminating the processing inserts. Flameproof light fixture conforming to Class IIA IIB of gas group shall be provided on the retarding chamber. c. The retarding chamber shall be sealed against the product container using pneumatically inflatable gaskets. d. Product Charging Port shall be provided for loading using pneumatic transfer system using suction of exhaust blower. e. A filter bags in P.C. Satin material shall be provided for very fine particles. f. An static dissipative bag of epitropic polyester material shall be provided for static sensitive products. g. The filter bag shall be sealed on the inside of the processor body using a pneumatically inflatable gasket. h. Explosion vent shall be provided.
5.	Bottom Chamber	This chamber shall be fabricated from S.S. 304 sheets. The sealing between bottom chamber and product container shall be done with pneumatically inflatable gaskets. The bottom chamber shall be radiused with drain from bottom for easy cleaning. The bottom chamber shall be fitted with a flange for connecting with AHU Supply duct. It shall be fitted with a butterfly valve ON/OFF type for isolating the machine from AHU when not in operation.

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6.	Air Handling Unit	Air Handling Unit shall be made from S.S. 304. Air Handling Unit shall be duly insulated with M.S. CRCA sheets. It shall consists of :- a. 10 Micron Coarse Filter (EU4), 5 micron Pre Filter (EU7) and 0.3 Micron HEPA Filter (EU13). All gaskets used will be in Silicon Rubber. The differential pressure across HEPA filter shall be indicated in the PLC. Magnehelic gauges shall be provided on 10 Micron and 5 Micron filters. b. Heating section shall house a steam radiator made from Stainless steel pipe and aluminum fins. The test pressure for steam radiator will be 200 PSIG. The inlet air temperature shall be controlled through PLC control system using PID Control System. c. It shall have ports for validation. • Steam supply Manifold and condensate return manifold shall be provided at an extra cost mentioned in price schedule. • Dehumidification Section can be provided at an extra cost.	
7.	Air Circulation	a. Exhaust Blower shall be driven by inverter based high efficiency 15 H.P. flameproof motor for drawing air through coater and enabling fluidization of particles. An inverter drive fitted to the blower motor shall control the fluidization velocity for desired batch size. (Final blower rating to be confirmed after site engineering is done on receipt of order) b. Required draught of air shall be produced using aluminum impeller housed in M.S. casing. Complete blower assembly shall be installed in service area or mezzanine floor. c. The exhaust blower shall be placed in service area. d. % Control Exhaust Air Damper shall be provided in the system for fine control of fluidization during process. Scrubber with outlet filter shall be installed at extra price.	

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8.	Heating	The air shall be heated with steam radiator, having steam consumption of 40.Kgs per hour The radiator shall be fabricated from S.S.304 quality materials and S.S.304 Fins and shall be hydrostatically tested at 200.PSIG. A PID –modulating steam control valve shall be provided loosely along with the machine, which shall have to be installed in steam supply line. The valve shall be of Normally close type and Air Fail to close type so that in case of utility failure it shall close automatically.
9.	<u>Spraying System</u>	a. Single Head type single arm spray nozzle is available for top spray granulation change part. b. Non-Pressurized solution feed vessel of 25 Liters with lid and pneumatically operated stirrer shall be provided. c. Flameproof Electrically operated peristaltic pump mounted on S.S. Trolley for fluid dosing to media nozzle via suitable tubing.
10.	Exhaust Air Police Filter Assembly	Exhaust Air Police Filter housing shall be made from painted Mild Steel. It shall house 10Micron Exhaust Air Filter assembly with Differential pressure indication on PLC for cleaning of exhaust air filter. Broken Bag Detector shall be installed Up Stream the Exhaust Filter Assembly to trip the system in case of any bag rupture or powder leakage through the system.
11.	<u>Safety Features</u>	The system shall have the following safety interlocks - Emergency Stop pressed - Earth Fault - Motor Overload - Inlet Valve Not Open - Compressed Air Pressure Low - Exhaust valve Not Open - Filter Not Up - Inflatable Gasket Pressure Low - Spray Gun Choke - System Temperature Very High The system shall create relevant Alarms and Trips in case of any of the above faults.

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10	Control	For Mitsubishi PLC with Color Touchscreen (Beijer/Mitsubishi). The PLC shall be able to control the entire process cycle along with recording on to a printer.(Online). The HMI shall be rated for installation into Zone 2 of hazardous area. The control system shall have the following features: 1. ON/OFF Button and Speed control for Blower Motor 2. PID Control for Steam Control Valve to control Inlet Air Temperature. 3. Indication of Product Temperature 4. Indication of Exhaust Air Temperature 5. Pneumatic Locking of Inflatable Seals for Product Container 6. Pneumatic Locking of Inflatable Seals for Filter Chamber 7. Automatic Shaking of Filter Bag 8. Online Printing 9. Differential Pressure Across HEPA Filter 10. Broken Bag Detector with analogue signal 11. Inlet Air CFM Sensor. M.S. Painted power panel shall be provided for housing the Power & Switchgear Components, Variable Frequency Drive (ABB), PLC (MITSUBISHI-Digital and Analogue Modules), MCSs (Siemens), SMPS- Omron, Relay cards, Terminals). The back side of the power panel shall house pneumatic components comprising of Pressure Switch, Solenoid Valves, Pressure Gauge etc. All pressure switches shall be of Danfoss Make, All pneumatic valves shall be of FESTO make. The Operating Panel shall be made from S.S. 304 with front door. It shall house the Touchscreen, Emergency Stop, Hooter and On/Off Button.	

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The control system shall have the following features:

- 1. ON/OFF Button and Speed control for Blower Motor
- 2. PID Control for Steam Control Valve to control Inlet Air Temperature.
- 3. Indication of Product Temperature
- 4. Indication of Exhaust Air Temperature
- 5. Pneumatic Locking of Inflatable Seals for Product Container
- 6. Pneumatic Locking of Inflatable Seals for Filter Chamber
- 7. Automatic Shaking of Filter Bag
- 8. Online Printing
- 9. Differential Pressure Across HEPA Filter
- 10. Broken Bag Detector with analogue signal
- 11. Inlet Air CFM Sensor.

M.S. Painted power panel shall be provided for housing the Power & Switchgear Components, Variable Frequency Drive (ABB), PLC (MITSUBISHI-Digital and Analogue Modules), MCSs (Siemens), SMPS- Omron, Relay cards, Terminals).

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PRICE : UNIFLUID 30- FLUID BED PROCESSOR

Sn.	Price	Item
1.	Rs. 45,00,000/-	For UNIFLUID 30 capacity GMP Model Fluid Bed Processor, having all contact parts made from S.S.316 quality materials, all non contact parts made from S.S.304 quality sheets, including the following: - PID Control Valve for Steam to control Inlet air Temperature 3 Stage Inlet Air Filtration Unit - Inlet ON/Off Valve - Exhaust % Control Valve. - Silicon Inflatable Gaskets - Dutch Weave Silicon Molded Mesh for Product Container - Product Sampler - Explosion Duct - Quick Remove Mesh Sensor for Inlet, Exhaust and Product Temperature Sensor. - Automatic Finger Bag Shaking. - Exhaust Air Blower with Flameproof Motor - Exhaust Air Police Filter Unit in S.S.304 - Mitsubishi PLC with 10" Touchscreen From Mitsubishi. - Inlet Air CFM Sensor - Inlet Air RH Indication - Broken Bag Detector(Analog type for easy set up)
2.	Rs. 5,50,000/-	For Extra Product Container including quick release mesh mechanism, product sampler, product temperature sensor made from S.S. 316 and Trolley made from S.S. 304.

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3.	Rs. 5,50,000/-	For Chilled Water based Dehumidifier section consisting of Aluminum fins and copper pipe heat exchange section, three way Chilled Water Control Valve (PID Dtype); Demister Pad, Drain Tray, along with U trap for condensate water.	
4.	Rs. 25,00,000/-	For DESSICANT TYPE DEHUMIDIFIER with Desiccant Rotor Wheel. The system shall comprise of Pre Cooling Coil Section, Desiccant Rotor Section, and Desiccant Regeneration Section. The Pre Cooling coils shall be made from Aluminum pipes and fins, The Desiccant Rotor system shall be regenerated using Steam and Electricity. The System shall be designed to work in two modes, basic dehumidification mode with only precooling coil and Heating Mode where the Rotor Desiccant shall dehumidify the air further as per the process requirement. Client shall be able to choose the mode from the HMI. NOTE: This dehumidifier shall be required if the product is sticky or moisture sensitive.	
5.	Rs. 8,500/-	For Polyester-Cotton Satin Bag	
6.	Rs. 20,500/-	For Polypropylene Antistatic Bag (Steel Wire Woven design)	
7.	Rs. 45,000/-	For Epitropic Polyester Filter Bag (With Woven Conducting Carbon Fiber design)	
8.	Rs. 1,65,000/-	For Valves and Fittings for Steam Inlet Line , comprising of 3. Nos Isolation Valve, Strainer OF THERMAX MAKE.	
9.	Rs. 1,25,000/-	For Condensate Return Line with Autodrain Valve, Steam Trap, Isolation Valve of THERMAX MAKE	
10.	Rs 75,000/-	For M.S. Silencer to be fitted downstream of blower.	

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11.	Rs. 7,50,000/-	For providing scrubber at the exhaust air side. It shall be made from M.S. and painted with epoxy paint. Pump shall be provided for recirculation of scrubbing solution to the scrubber.
12.	Rs. 5,50,000/-	WIP Nozzle System. The WIP System shall comprise of spray nozzles located at 3 strategic locations in the Fluid Bed System. High Pressure Braided hose shall be provided for connection with the Sub headers and main header. Drain Valve shall be provided in the bottom chamber for connection of Plastic Braided Hose Pipe to the drain. High Pressure Pump Included & Piping in Client Scope.

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Ducts and Bends (To be finalized as per the layout)

1.	Rs. 25,000/-	Extra for each meter length of straight duct fabricated from S.S. 304 quality material.
2.	Rs. 28,500/-	Extra for each 90Deg elbow fabricated from S.S. 304 quality material
3.	Rs. 52,000/-	Extra for each meter length of insulated duct fabricated from S.S. 304 quality material.
4.	Rs. 60,000/-	Extra for each insulated elbow fabricated from S.S. 304 quality steel
5.	Rs. 35,000/-	For each meter length of Explosion Flap Duct made from S.S. 304.

INSTALLATION & COMMISSIONING & VALIDATION

6.	Rs. 1,50,000/-	For Installation & Commissioning: CLIENT'S SCOPE OF SUPPLY: 1. Level floor. 2. Foundation and necessary masonry work if required. 3. Shifting of equipment at site. 4. Steam connection along with necessary steam fittings such as steam trap, strainer etc. and steam controlling device. 5. Necessary unskilled help at your cost. 6. All the necessary electrical cables from the Clients MCCB to the individual equipment shall be in clients scope of supply 7. All pneumatic tubes to and fro from Power Panel to the machine shall be in client's scope of supply 8. All electrical power and signal cables from Clients MCC to the Individual equipment Power Panel/Operating Panel shall be in clients scope of supply. 9. Cables Trays and Routing of Cable Trays
7.	Rs. 55,000/-	For Validation Documents Comprising of DQ, IQ and OQ
8.	Rs. 2,50,000/-	For all power and signal cables along with pneumatic tubes for the machine. Client shall have to bring in the main power cable to the power panel.

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2. TERMS & CONDITIONS:

Prices : Prices offered are net-ex-works Mumbai exclusive of Packing, Carting, Crating, Forwarding, Freight and any other incidental charges will be extra to your account.

Taxes : GST shall be charged extra as and if applicable at the time of dispatch.

Packing & Forwarding: 2.5% EXTRA

Inspection : Final Inspection prior to dispatch will have to be carried out jointly by us and your representative at our Mumbai works only. You will be intimated as soon as the machines are ready for dispatch.

Transport : Shall be paid by you directly to the transporter.

Delivery : Within 24 WEEKS.

Payment : 1. 40% Advance on confirmation of order.
2. 50% Balance against proforma invoice after FAT
3. 10% After Installation and Commissioning. (Not exceeding 1 months after supply)

Validity : This offer is valid for 30 days.

Warranty : We undertake to make good by replacement or repair, defects arising out of faulty design, materials or workmanship within 12 months of the date of dispatch, purely on ex-works basis i.e., the defective part will be returned by the buyer to our works and replacement taken by the buyer from our works. Bought-out components are guaranteed by us only to the extent of guarantees given to us by our suppliers. Electrical components, such as heaters, motors, contactors etc. rubber components and instruments such as pressure gauges, thermometers, etc. and other such consumables, etc. are however not covered under this Warranty. Warranty stands void if spare parts other than OEM are used.

Jurisdiction : Any dispute arising out of this quotation is subject to the **JURISDICTION OF MUMBAI COURT ONLY**. All dimensions are approximate.

Design and Specification of all our products are subject to alteration, without notice.

Thanking you and awaiting for your favorable reply.

Yours truly,

For HUGOPHARM TECHNOLOGIES PVT LTD.
Hiren Panchal